1. **Install the Print Server role on your Windows Server and add a new network printer using the Print Management console.**

Part 1: Installing the Print Server Role

1. **Open Server Manager:**
   - Log in to your Windows Server and open Server Manager (by default yeh startup par open hota hai).
2. **Add Roles and Features:**
   - Dashboard par click karein Add roles and features.
3. **Before You Begin Screen:**
   - Click Next (aap *Skip this page by default* bhi select kar sakte hain).
4. **Select Installation Type:**
   - Choose Role-based or feature-based installation and click Next.
5. **Select Destination Server:**
   - Select your server from the server pool (it is selected **by** default) and click Next.
6. **Select Server Roles:**
   - Scroll down the list and check the box for Print and Document Services. ◦ A pop-up window will appear asking to add required features; click Add Features, then click Next.
7. **Select Role Services:**
   - Ensure that Print Server is checked (checked by default). You can also select *LPD Service* if legacy Unix printing is required, then click Next.
8. **Confirmation & Installation:**
   - Review your selections and click Install.
   - Wait for the installation to finish, then click Close.

Part 2: Adding a Network Printer via Print Management Console

1. **Open Print Management Console:**
   - Press Win + R, type printmanagement.msc, and press Enter (or open it from Server Manager -> Tools -> Print Management).
2. **Navigate to Printers:**
   - In the left panel, expand Print Servers, then click on your local server name.
   - Click on Printers in the expanded tree.
3. **Launch Add Printer Wizard:**

- Right-click anywhere in the middle blank pane (or on *Printers*) and select Add Printer...

4. **Configure Printer Connection:**
   - Select Add a TCP/IP or Web Services printer by IP address or hostname and click Next.
5. **Enter Printer Details:**
   - Select TCP/IP Device as the Device type.
   - Enter the printer's IP Address or Hostname (e.g., 192.168.1.200) and click Next. The wizard will detect the network printer**.**
6. **Install Driver & Finish:**
   - Select an appropriate printer driver from the list (or browse to install a manufacturer driver). Click Next.
   - Provide a printer name and share name (if you want to share it on the network), then click Next and Finish.

2. **Configure print permissions for a shared network printer so that only specific users (create two test user accounts) can print, while others are denied access.

Hint: Use the Security tab in the printer's Properties to set Allow/Deny permissions for each user.** ⭘

Part 1: Create Two Test User Accounts

1. Press Win + R, type lusrmgr.msc, and press Enter to open *Local Users and Groups*.
2. Right-click on Users in the left panel and select New User...
3. Create the first test user:
   - Username: TestUser1 - Set a password, uncheck *User must change password at next logon*, and check Password never expires.
   - Click Create.
4. Create the second test user:
   - Username: TestUser2
   - Set a password, uncheck *User must change password at next logon*, and check Password never expires.
   - Click Create, then click Close.

Part 2: Configure Printer Security Permissions

1. Press Win + R, type printmanagement.msc, and press Enter to open the *Print Management* console (or open *Devices and Printers* via Control Panel).
2. Expand Print Servers -> [Your Server Name] -> Printers.
3. Right-click your network printer and select Properties...
4. Switch to the Security tab.

Access Control Precaution

Be careful when removing default groups like Everyone. Ensure your administrator account retains Manage Printer permissions to avoid locking yourself out of configuring the printer later.

5. Restrict Public Access:
    - Select the Everyone group from the *Group or user names* list.
    - Click Remove (or under permissions for *Everyone*, check Deny for Print).
6. Grant Access to Specific Test Users:
    - Click Add...
    - Type TestUser1; TestUser2 in the object names box, click Check Names, and press OK.
    - Select TestUser1 in the list and ensure Allow is checked for Print.
    - Select TestUser2 in the list and ensure Allow is checked for Print.
7. Click Apply and OK to save the permission settings.

Part 3: Testing & Verification

User Account	**Action Performed**	**Result / Status**
TestUser1 / TestUser2	Send a test page print job to the shared network printer.	Success (Print job reaches queue and prints).
Any Other User / Guest	Attempt to connect or send a print job to the printer.	Failed (Access Denied / Unable to connect).

3. **Send a test print job from both an allowed and a denied user account, then use the Print Management console to monitor and manage the print queue, noting what happens for each user.**

Procedure & Documentation:

Step 1: Sending Print Jobs from Client Systems

1. From the Allowed User (TestUser1):
    - Log in to a client machine using TestUser1 credentials. Open any document or text file, press Ctrl + P, select the shared network printer, and click Print.
2. From a Denied User (e.g., standard user / guest):
    - Log in using an account that was not granted Print permissions.
    - Try to send a print job to the same shared network printer.

Step 2: Monitoring the Print Queue via Print Management 1. On your Windows Server, open the Print Management console (printmanagement.msc).

2. Navigate to Print Servers -> [Your Server Name] -> Printers.
3. Right-click your shared network printer and select Open Print Queue to monitor incoming jobs in real time.

Observations & Results:

User Type	**Action / Print Job Status**	**Print Management Queue Result**
Allowed User (TestUser1)	Sent a test document to the printer.	Success: The job appears in the print queue momentarily with status *Spooling / Printing* and successfully processes to the physical or virtual printer.

Denied User	Attempted to send a document or connect	Blocked / Failed: The print job is immediately rejected, and an "Access Denied"
	to the printer.	error message pops up on the client screen. No job appears in the server's print queue.

How to Manage Print Jobs in the Queue (Administrative Actions):

- Pause / Resume: If you need to stop printing temporarily, right-click the printing document in the queue and select Pause. Select Resume to restart.
- Cancel / Delete a Job: If a job is stuck or needs to be removed, rightclick the document and select Cancel (or *Delete*), which instantly clears it from the queue.

4. **Change the priority of two print jobs in the print queue so that one user's job prints before another's, regardless of submission order.

Hint: Right-click the job in the print queue and adjust the priority value.**

Procedure & Documentation:

Step 1: Open the Print Queue

1. Apne Windows Server par Print Management console kholein (printmanagement.msc run karke).
2. Left side pane se Print Servers -> [Your Server Name] -> Printers par jayein.
3. Apne shared network printer par right-click karein aur Open Print Queue select karein. (Isme aapko current printing jobs dikh jayengi jo alag-alag users ne bheji hain).

Step 2: Adjust Job Priority

1. Print queue list mein us job ko dhundhein jise aap pehle print karwana chahte hain (chahe woh baad mein submit ki gayi ho).
2. Us job par right-click karein aur Properties par click karein.
3. General tab ke andar Priority slider ya number (jo ki default 1 se 99 ke beech hota hai) dikhega.

- *Note:* Higher number ka matlab hota hai higher priority (jaise priority ko badhakar 99 kar dein).

4. Doosri job (jise baad mein print karwana hai) ki priority ko kam rakhne dein (jaise default 1 ya 5).
5. Apply aur OK par click karein**.**

5. **Simulate a real-world scenario: Imagine your college fest team is printing event posters. Configure the printer so that 'Event Managers' group can manage documents (pause, resume, cancel), while 'Volunteers' can only print, and test these permissions by logging in as both role**

Step 1: Create the User Groups (Prerequisite)

1. Apne Windows Server par Win + R press karke lusrmgr.msc type karein aur Enter dabayein.
2. Left panel mein Groups par click karein, right-click karke do naye groups banayein: Event Managers Volunteers
3. In groups ke andar apne test users (jaise *ManagerUser* aur *VolunteerUser*) ko add kar dein.

Step 2: Configure Printer Permissions in Print Management

1. Press Win + R, type printmanagement.msc, and press Enter to open the *Print Management* console.
2. Go to Print Servers -> [Your Server Name] -> Printers.
3. Right-click your poster printer and select Properties...
4. Switch to the Security tab.
5. Assign Permissions for 'Volunteers':
   - Click Add..., type Volunteers, click Check Names, and press OK.
   - Select Volunteers in the list and check Allow for Print only (ensure *Manage Documents* and *Manage Printers* are unchecked).

6. Assign Permissions for 'Event Managers':
   - Click Add..., type Event Managers, click Check Names, and press OK.
   - Select Event Managers in the list and check Allow for both Print and Manage Documents. *(Manage Documents permission automatically allows pausing, resuming, restarting, and deleting any print jobs in the queue).*
7. Click Apply and OK to save the settings. **Step-by-Step Testing & Observation:**

Role / User Logged In	Action Performed	Result / Behavior
Volunteer (VolunteerUser)	**Sends a poster print**	**Partial Access: Poster prints**
	job, then tries to pause or cancel someone else's job in the queue.	**successfully, but when attempting to pause/cancel a job, an Access Denied error appears because they only have *Print* rights.**
Event Manager (ManagerUser)	**Sends a poster print job, and opens the printer queue to manage documents.**	**Full Control of Queue: Can successfully send jobs, and has full authority to Pause, Resume, Restart, or Cancel any poster job currently waiting in the queue.**